

UERANSIM + Open5GS Network Performance Documentation

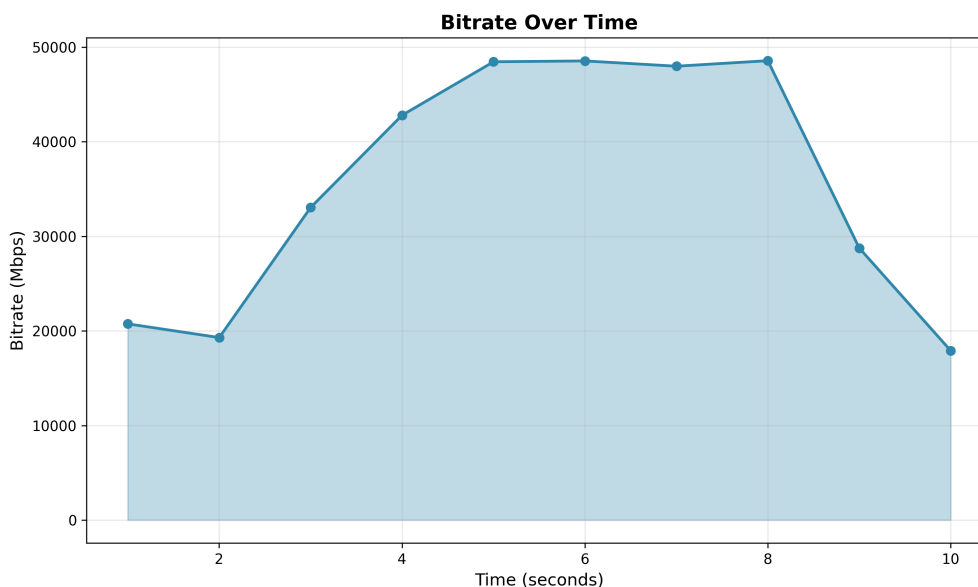
1. Working Baseline Topology + Diagram



2. Baseline Measurement Results

Performance Metrics

- **Bitrate (Mbps) - 9.99; Jitter (ms) - 0.010; Avg RTT (ms) - 0.327**



Overhead for IPv4

- GTP-U: $20(\text{IP}) + 8(\text{UDP}) + 12(\text{GTP-U}) = 40$ Bytes
- GTP-U with IPsec: $20(\text{IP}) + 8(\text{UDP}) + 12(\text{GTP-U}) + 58(\text{IPsec}) = 98$ Bytes
- QUICUP: $20(\text{IP}) + 8(\text{UDP}) + 34(\text{QUIC/Auth Tag}) = 62$ Bytes

Approx. 20 bytes more for IPv6

3. QUIC-Based User-Plane Architecture



Implementation Approach:

1. **Direct Replacement:** Replace GTP-U with QUIC streams in the N3 interface
 - Simplified architecture
 - Reduced protocol overhead
 - Requires modifications to both UERANSIM and OPEN5GS

2. **Encapsulation Mode:** Encapsulate GTP-U within QUIC tunnel

- Maintains 3GPP compatibility
- Easier integration with existing infrastructure
- Gradual migration path

4. QUIC Library Evaluation

Library	Language	License	Features
quiche (Cloudflare)	Rust (C/C++ bindings)	BSD-2-Clause	Production-ready, HTTP/3 support
msquic (Microsoft)	C	MIT	IETF RFC 9000, Windows/Azure proven
ngtcp2 (nghttp2)	C	MIT	Lightweight, minimal dependencies
lsquic (LiteSpeed)	C	MIT	High performance, web-optimized
mvfst (Meta)	C++	MIT	Meta-scale tested, advanced congestion control

Recommended Implementation: **msquic**

- **Native C API:** Seamless integration with Open5GS C codebase
- **Cross-platform:** Linux compatibility for 5G deployments
- **License compatibility:** MIT license suitable for open-source projects